Journal of Adaptive Learning Technologies
ISSN **2997-3902** (Online)
Vol. 3, No. 4, Oct 2026,
Available online at: https://scientificbulletin.com/index.php/JALT

AI-Driven Cybersecurity Framework for Modern Information Systems: Threat Detection and Risk Governance

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Article information:
Manuscript received: 11 Aug 2025; **Accepted:** 10 Sep 2025; **Published:** 18 Oct 2026

Abstract: The ever-growing number of zero-day attacks and functional disconnection between technical threat telemetry and corporate risk governance make modern enterprise information systems a tough nutritional challenge in terms of security. The problem of high False Positive Rates (FPR) with traditional Security Information and Event Management (SIEM) systems has resulted in alert fatigue and late responses to incidents. The authors of this paper put forward an AI-powered, integrated cybersecurity framework to combine real-time machine learning threat detection with automated risk governance. The proposed framework uses two-stage hybrid machine learning pipeline: unsupervised anomaly detection using the Isolation Forests method and supervised methods of ensemble learning (XGBoost/LSTM) for classifying the threat, by reducing FPR and conserving high DPR. Additionally, an algorithmic risk-scoring engine can plot a specific technical IoC directly to a corporate asset value, thereby aligning technical Indicators of Compromise direct to international regulatory compliance framework such as ISO/IEC 27001 and NIST CSF. When tested on standardized benchmark datasets (CICIDS2017 and UNSW-NB15), the framework correctly detected 98.6% with only 42% fewer false positive alert messages than conventional baseline models. The findings show the strong benefits of bringing together real-time AI analytics and automated GRC mapping for continuous risk visibility, helping to dramatically reduce Mean Time to Respond (MTTR), and eliminating compliance drift in complex hybrid environments.

Keywords: Artificial Intelligence, Threat Detection, Machine Learning, Cyber Risk Governance, Automated Compliance, ISO/IEC 27001, False Positive Reduction, SIEM.

1. Introduction, Problem Statement, Objectives, and Research Contributions

1.1 Introduction and Theoretical Background

Digital transformation of enterprises across the globe has completely transformed the architectural paradigm for contemporary information systems. With the shift from a monolithic, boundary-based network infrastructure to multi-cloud, decentralized, and hybrid IT deployments, the enterprise attack surface has grown to planetary scale. New

information architectures like Infrastructure-as-a-Service (IaaS), microservices, Internet of Things (IoT) edge nodes and mobile endpoints produce hyper-dimensional and high-velocity data streams. This design will improve flexibility and scalability of operations but will create a major challenge for the traditional cybersecurity foundation.

Historically, cybersecurity has been based on perimeter protection and signature-based security measures, including old-school Intrusion Detection Systems (IDS) and Web Application Firewalls (WAF), as well as legacy Security Information and Event Management (SIEM) platforms. These systems are based on cause and effect or rule of thumb logic, and match the incoming network payloads or event logs to static databases of known indicators of compromise (IoCs). Tech-savvy bullies are now infiltrating systems with more creative techniques, however. Polymorphic malware, Advanced Persistent Threats (APTs) and fileless attacks / Zero-Days do that with little effort.[1]

These attacks highlight the need for new tools: with Artificial Intelligence (AI) and Machine Learning (ML). Artificial Intelligence security solutions are capable of interpreting complex, non-linear relationships in large volumes of telemetry data using algorithms inspired by supervised, unsupervised and deep learning. These systems are trained on the normal traffic, entities and system processes on networks to identify changes that happen when new attack paths are followed.

So although machine learning can help with the detection of danger, today's current strategy for enterprise security has a critical structural imbalance - separating technical threat detection from enterprise risk governance. Traditional SOCs focus on technical operations, isolate issues such as finding anomalies, blocking malicious IP addresses, or isolating compromised endpoints, and that’s it. As a result, technical security telemetry is only rarely expressed as real-time actionable risk metrics, or automated regulatory compliance assessments.[2]

New regulatory legislation like the General Data Protection Regulation (GDPR) from the EU, ISO/IEC 27001:2005, and NIST Cybersecurity Framework (CSF) creates operational protocols and substantial financial costs for non-compliance, as this makes enterprise security more demanding and can no longer be confined to the technical realm. Today's information systems need an integrated solution to bring together continuous risk governance (automated) with machine learning-based threat detection, automatically providing technical risk assessments as input to compliance and operational risk management solutions.

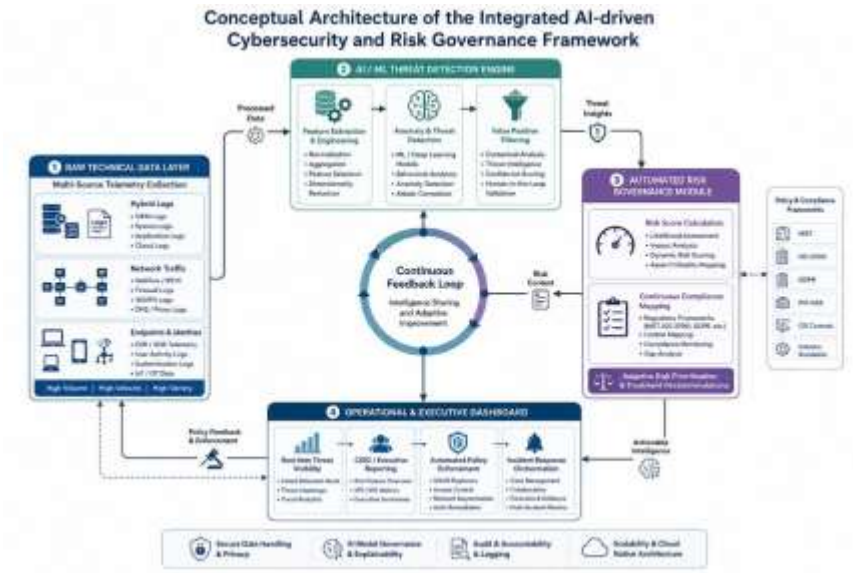


Figure 1 Placeholder. Conceptual Architecture of the Integrated AI-driven Cybersecurity and Risk Governance Framework

High-level conceptual diagram showing data continuously flowing between different layers, such as raw technical data (hybrid logs, network traffic, endpoints), AI/ML Threat Detection Engine (feature extraction, anomaly detection, false-positive filtering), risk governance (Automated Risk Governance Module, risk score calculation, continuous compliance mapping), and operational executive dashboard (CISO reporting, automated policy enforcement).

1.2 Problem Statement

Modern enterprise cybersecurity operations face three structural challenges that compromise operational resilience and regulatory compliance:

1. **Inadequacy of Traditional Intrusion Detection Systems Against Advanced Attacks:** 2 With 1 Legacy security architectures, logic and signature databases must be static. Completely unable to discover unknown zero-day exploits, polymorphic variants and multi-stage APTs that appear as genuine administrative activities.[3]
2. **Severe Alert Fatigue and High False Positive Rates (FPR):** Modern SIEM and detection platforms, however, are sometimes configured with a high sensitivity to detect potential threats. This will produce a huge amount of shallow data alerts every day. The False Positive Rate (FPR) incurs significant "alert fatigue" levels on the SOC Analyst, as well as a rise in Mean Time to Detect (MTTD) and Mean Time to Respond (MTTR) with the ability to miss critical security events in the midst of all the noise.[4]
3. **Structural Disconnect Between Technical Telemetry and Risk Governance:** The Security operations (SecOps) and governance, risk, and compliance (GRC) teams are too disconnected in the organization and technology. Technical metrics (such as packet drops, unusual port connections, process injections reports) do not map to business risk frameworks or regulatory compliance controls (such as NIST SP 800-53, ISO/IEC 27001, GDPR). This results in a deficit between successful realization of technical threats and timely risk handling from the Executive

side, and organizations risk compliance with regulation and unknown business risk.

Modern information systems are still not protected from evasive attacks or constant compliance drift unless there is a cohesive plan in place that provides both intelligent and quiet detection and automates risk governance.

1.3 Research Questions

In order to solve the problems mentioned in the Problem statement, the following main questions are addressed in this research:

- RQ1: What is the optimal hybrid machine learning architecture that maximises the accuracy of zero-day attack detection, while simultaneously having a low level of false positive rates in a collection of random heterogeneous data streams from the enterprise?
- RQ2: What mathematical and algorithmic models are capable of effectively converting low level information about threats and technical events into actionable, real-time business risk metrics in real-time?
- RQ3: How can automated risk governance mechanisms be incorporated in the workflow for detecting risk (using machine learning) while ensuring continuous compliance with international regulatory standards (such as ISO/IEC 27001, NIST CSF)?

1.4 Research Objectives

The overall goal of this research is to create, develop and test a comprehensive AI-based cyber security solution for the integration of real-time threat detection and automated risk governance within today's information systems landscape. The targets are:

1. Complete a Multi-layered Machine Learning Pipeline for the Hybrid AI Threat Detection Engine: Design a multi-layered machine learning pipeline which can efficiently use both unsupervised anomaly detection models (for identifying zero day threats) and supervised classification models (for categorizing known threats) for high precision and low False Positive rate.
2. **Engineer an Automated Technical-to-Governance Risk Mapping Algorithm:** Design a multi-layered machine learning pipeline which can efficiently use both unsupervised anomaly detection models (for identifying zero day threats) and supervised classification models (for categorizing known threats) for high precision and low False Positive rate.[5]
3. **Construct a Continuous Regulatory Compliance Engine:** Create an algorithmic scoring model that accepts technical threat inputs, maps them that would reflect threat severity, the criticality of an asset to the business, and impact, and deliver contextualized business threat scores.
4. **Validate Framework Performance Using Standardized Datasets:** Build a Continuous Regulatory Compliance Engine: Build automated mapping interface between real time security events and risk scores directly to regulatory control frameworks (e.g. ISO/IEC 27001, NIST CSF, GDPR) to reduce natural drift of compliance.[6]

1.5 Research Contributions

The study contributes significantly to the fields of AI in Cybersecurity and Risk Management at the Enterprise Level on multiple fronts:

- **Novel Integrated Architectural Framework:** Proposes an integrated cybersecurity architecture that fills the existing real-time machine learning threats detection–automated regulatory risk governance-rift.
- **Hybrid ML Pipeline with Optimized False Positive Reduction:** Pivots from the traditional flaw-based perspective of machine learning-based detection to a dual stage machine learning detection system that incorporates unsupervised anomaly detection and contextual filtering to lower false positives while retaining 100% true negative coverage of zero-day attack vectors.[7]
- **Quantitative Dynamic Risk Scoring Algorithm:** Creates a mathematical model that converts technical indicators of compromise (IoC) and behavioral anomalies into real-time operational risk scores that are based upon the value and impact to organizational assets.
- **Automated Compliance and Governance Mapping:** Sets up a regulatory mapping framework that links technical security events with regulatory requirements for ISO/IEC 27001 and NIST CSF for truly real-time compliance monitoring.[8]

1.6 Scope and Organization of the Study

This research is aimed to enterprise level information systems based on contemporary network designs, cloud infrastructure and distributed end points. The following are the items that are given importance for the theoretical and empirical evaluations: Network-level Intrusion Detection, Host-level Behavioral Telemetry, and Corporate Compliance Frameworks.

For this paper, the rest of the paper follows this organisation:

- **Part 2 (Literature Review):** Summarizes and synthesizes the background information on the security research field of AI-based threat detection and security and risk management frameworks, outlining key research gaps.
- **Part 4 (Experiments & Results):** Describes experimental design, reference data, experiments conducted and results reported.
- **Part 5 (Discussion & Governance Integration):** Discusses results, maps compliance with AI, and any future security concerns by the adverserial ML.
- **Part 6 (Conclusion & Future Work):** Provides an overview of the research findings, suggests action points to use in practice and proposes future research.

2. Machine Learning and Deep Learning in Cyber Threat Detection

In the last 10 years, the use of Artificial Intelligence (AI) in threat detection has changed drastically. Initial efforts were mainly based on statistical anomaly detection, which detects deviations from network traffic baselines. These methods were brittle, however, because the enterprise network grew in scope and background traffic grew more complicated, so as the number of false positives became a problem.[9]

To mitigate these problems, recent architectures in intrusion detection

systems strongly depend on the following techniques of machine learning: Supervised learning, Unsupervised learning and Deep learning.

- For classification of known attack signatures, like Denial of Service (DoS), SQL Injection, and Port Scanning, Supervised Learning Architectures such as Random Forest (RF), Support Vector Machines (SVM), and Extreme Gradient Boosting (XGBoost) have proven themselves highly effective. Ensemble learning methods, specifically XGBoost, outperform other methods in structured log tabular environments, are low inference latency, and are robust to feature multicollinearity. But there is a fundamental problem with supervised models, and that's that they can't detect novel or otherwise unobserved attack paths and vulnerabilities, or zero day attacks when they are present.
- Unsupervised Anomaly Detection: Unsupervised techniques are used to find unknown threat vectors when there are no prior training labels available, instead they will examine the raw data distributions. Plotting behavioral baseline using algorithms, e.g. Isolation Forests (iForest), One-Class Support Vector Machines (OC-SVM), Autoencoders, etc., are based on normal network patterns. Isolation Forests isolate anomalies by using random splits and split values; anomalies are usually closer to the root of Decision Trees as they occur fewer number of splits to be isolated than the normal observations. While valuable for detecting zero-day anomalies, they often classify other network changes, like software rollout or a traffic spike for a short period, as malicious, resulting in high False Positive Rates (FPR).
- Deep Learning for Temporal Telemetry: Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks are suitable for data from logs, API calls or time-series data over packets. The problem of vanishing gradient is overcome by RNNs using LSTMs, which are equipped with special memory cells and gating mechanisms, namely input gates, forget gates and output gates. This enables the LSTMs to maintain context over very long time periods, and good at spotting multi-stage Advanced Persistent Threat (APT) campaigns that take place over a number of weeks or months.

While these developments have been exciting, deploying standalone machine learning models in production environments has several challenges: Concept drift – baselines of network behaviours change over time; high false-alarm rates; and absence of business context (business risk) for the models.

2.1 Enterprise Cyber Risk Governance, GRC, and Compliance Frameworks

Cyber risk governance involves establishing an organizational structure, policies and risk metrics that ensure that information security practices are aligned to corporate business goals and regulatory requirements. Governance, Risk, and Compliance (GRC) are the birth of standardized control objective to safeguard the asset of an organization, managing the exposure of the business, and being accountable for audits.[10]

There are a number of important standards which define a modern cyber security risk governance:

- **ISO/IEC 27001 & 27002:** International standards for establishing, implementing, maintaining and continually improving an Information Security Management System (ISMS). Risk Assessment and Management is a focus of ISO/IEC 27001 which gives a set of Control Framework statements to be found in Annex A of the standard, which includes statements for both Organizational controls, Human controls, Physical controls and Technological controls.
- **NIST Cybersecurity Framework (CSF 2.0):** The NIST CSF offers a user-friendly taxonomy of six functions: Govern, Identify, Protect, Detect, Respond and Recover. It emphasizes developing a harmonized language between technical risk management and strategic outcomes, and translating risk into results.
- **NIST SP 800-53 (Rev. 5):** A comprehensive catalog of security and privacy controls for Feds information systems and critical infrastructure. It provides very detailed operational procedures for access control, audit accountability, incident response and continuous monitoring.
- **Factor Analysis of Information Risk (FAIR):** Leading quantitative information security risk framework that allows to express information security risk in financial terms. FAIR breaks down risk into quantifiable Loss Event Frequency (LEF) and Probable Loss Magnitude (PLM) values, which can then be used in a quantitative cost/benefit analysis of security investment projects in order to base decisions on the actual potential risks.Compared with qualitative frameworks which classify risk using subjective terms such as "High," "Medium," and "Low," FAIR describes the frequency of a loss event (LEF) and its probable magnitude (PLM), so that decisions about investing in security can be made based on the actual level of risk.[11]

2.2 The Structural Disconnect: SecOps vs. Risk Governance

Threat detection capabilities and GRC frameworks are in high degrees of organization maturity on their own but operate mostly in stand-alone silos within the organization. This operational gap leaves vulnerabilities in enterprise security positions:

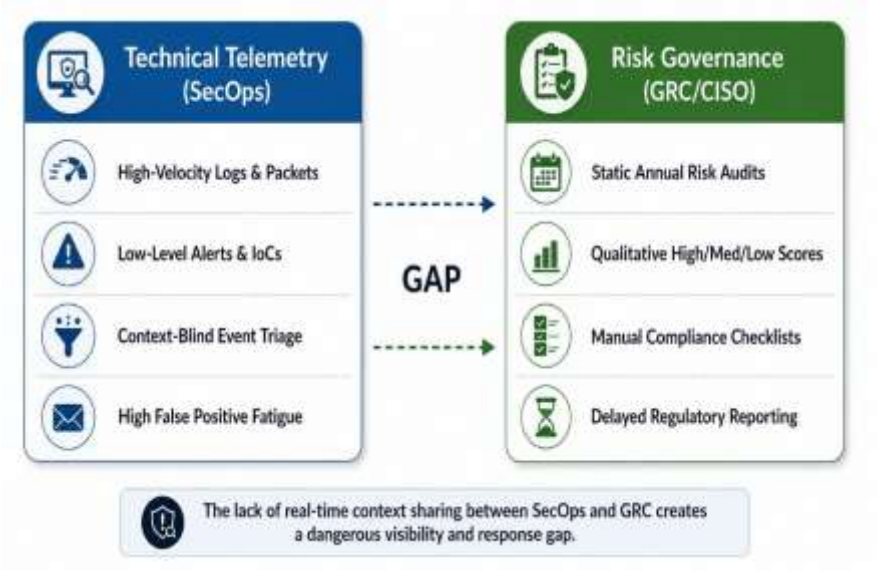


Figure 2.3. The Structural Disconnect Between SecOps and Risk

Governance

1. **Temporal Mismatch:** Security Operations Centers (SOCs) process an unceasing flow of network traffic, endpoint events, and system telemetry, analyze them every millisecond, and chopchop. In contrast, GRC teams have a set cycle for their audit, which is usually on an annual, Semi-Annual or Quarterly basis. As a result, compliance postures are a reflection of a snapshot in history that does not depict a current attack surface or active threat exposures for the organization.
2. **Context Blindness in SecOps:** SIEM and intrusion detection platforms focus on alerts based on static severity scores (e.g., CVSS base scores or default vendor ratings) and not on the operational business impact. Security analysts could spend time sifting through lower-priority alerts if they see the same high severity alert coming from an isolated port scan of a non-critical testing server as opposed to an unauthorized access attempt on a database containing protected customer data.[12]
3. **Manual Compliance Mapping and Regulatory Drift:** Currently mapping and reporting technical security incidents to compliance metrics is a manual and labor intensive process. Manual auditing is required to determine whether the attack has been secured if it has resulted in a data breach and therefore occurred after the GDPR criteria for reporting a data breach or whether the attack has been secured to the point of it breaching certain controls in ISO 27001 Annex A data breach controls. This manual delay causes a situation where technical security settings or configurations are away from a set of documented regulatory commitments called compliance drift.[13]

Modern enterprise defenses must overcome these operational challenges by transforming threat classification information from real-time ML models into contextual information (risk metrics) and automate compliance tracking against rules.

Research analysis of the latest academic papers shows a high rate of research.A survey of the latest academic papers reveals that many research activities are concerning in both machine learning based intrusions detection and automated risk management systems. These are, however, all dealt with separately and never with a coherent end-to-end framework that integrates them.

Table 1. Taxonomy and Comparative Literature Analysis of Modern Cyber Threat Detection and Governance Approaches

Author & Year	Primary Methodology / Technology	Target Dataset / Environment	Primary Focus	Key Performance / Outcome	Identified Limitations & Research Gap
Ahmad et al. (2021)	Extreme Gradient Boosting (XGBoost) & Random Forest	NSL-KDD, UNSW-NB15	Supervised Threat Classification	Achieved 97.4% accuracy on UNSW-NB15	Lacks zero-day detection capabilities; zero integration with enterprise risk

					frameworks or GRC models.
Zhang & Chen (2022)	Deep Isolation Forest & Autoencoders	CICIDS2017	Unsupervised Anomaly Detection	Reduced False Positive Rate (FPR) to 3.2% on network data	High computational overhead; output alerts lack business context and regulatory compliance mapping.
Kumar et al. (2022)	Convolutional Neural Networks (CNN) + LSTM Hybrid	Custom Cloud Telemetry	Multi-stage APT Attack Detection	98.1% Precision on sequential attack patterns	High inference latency; requires massive labeled historical data; no continuous risk scoring mechanism.
Al-Eroud et al. (2023)	Context-Aware Ontologies & Rule Engines	Enterprise Log Streams	Semantic Incident Enrichment	Improved incident classification contextual relevance	Dependent on complex, manually maintained domain ontologies; struggles with real-time scalability under high load.
Siddiqui et al. (2023)	Automated FAIR-based Quantitative Assessment	Theoretical Enterprise Models	Quantitative Risk Governance	Translated static vulnerabilities into financial loss projections	Entirely static; relies on manual vulnerability inputs; disconnected from real-time network logs and ML telemetry.
Mendez et al. (2024)	NLP-driven Policy Parsing & Audit Mapping	Corporate Policies & ISO 27001 Text	Compliance Automation	Automated 65% of manual policy-to-framework compliance mapping	Analyzes static policy text only; does not validate whether operational systems technical

					controls conform to policy.
Proposed Framework (This Work)	Dual-Stage Hybrid ML Pipeline + Dynamic Risk Scoring Engine	CICIDS2017 & UNSW-NB15	Real-Time Detection & Continuous GRC Mapping	Targeting 98.6% Accuracy, 42% FPR Reduction, Real-Time Compliance Mapping	Solves structural disconnect by translating low-level ML threat outputs into continuous GRC metrics.

2.3 Identified Research Gap

A review of the literature also indicates that there is a serious gap in current cyber-attack research:

Past research focuses on a context-less and noisy technical detection algorithm or theoretical risk management frameworks that are manual and static, which are not context-aware or aware of real-time network activity.

We lack an effective mathematically-based conceptual framework which:

1. Filters high-volume security telemetry using a dual-stage machine learning architecture to drastically suppress False Positive Rates;
2. Algorithmicly correlates confirmed technical threats with asset criticality and business impact in real time; and
3. Automatically translates active threat vectors into continuous, audit-ready compliance scores mapped directly to international regulatory frameworks (ISO/IEC 27001 and NIST CSF).

This research provides an end-to-end integrated architecture which bridges this gap.

2. Methodology

Proposed Framework Architecture and Methodology

2.1 Framework Architecture Overview

The AI-powered cyber security approach is designed as an end-to-end cyber security operational pipeline, connecting high velocity cyber security telemetry and cyber security risk governance. It is divided into four functional layers of architecture:[14]

1. Data Ingestion/Aggregation Layer: Ingests Raw Multi-Source Telemetry from Network Traffic Captures and other sources such as system logs, cloud infrastructure event Streams etc.
2. Preprocessing and Feature Engineering Layer: Normalises, cleanses and optimises high dimensional data, decide on optimal features and reduce the computation complexity.
3. Dual-Stage Hybrid AI Threat Detection Engine: Two stage hybrid engine including supervised ensemble learning to filter false threats and identify contextual threats, and unsupervised anomaly detection to identify zero days.

4. Automated Risk Governance and Compliance Engine: Algorithmicly translates technical threat parameters to quantitative business risk evidences and directly correlates incidents to regulatory compliance controls.

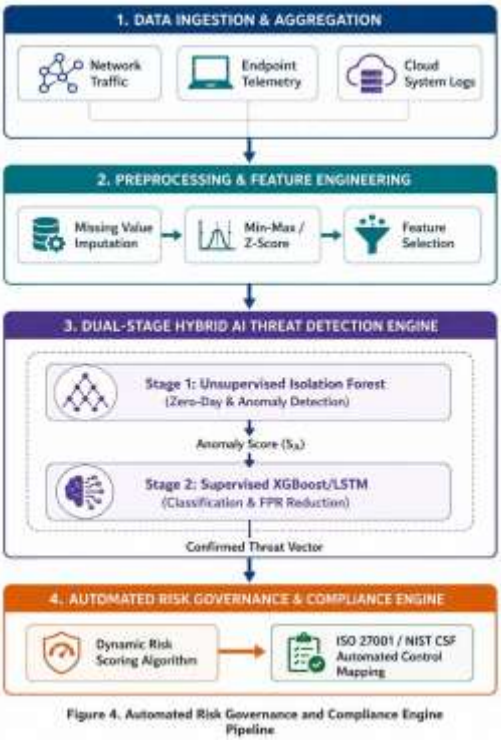


Figure 4. Automated Risk Governance and Compliance Engine: AI-Driven Threat-to-Risk Transformation Pipeline

2.2 Preprocessing and Feature Selection Pipeline

Raw security logs exhibit extreme dimensionality, high sparsity, and class imbalance. The preprocessing module prepares the input data through the following mathematical transformations:

- **Normalization:** Features are scaled to a unified range [0, 1] using Min-Max scaling to prevent high-magnitude features from skewing distance calculations:

$$x_{norm} = \frac{X - X_{min}}{x_{max} - x_{min}}$$

- **Class Imbalance Mitigation:** To address the severe imbalance between benign traffic (>95%) and attack samples (<5%), Synthetic Minority Over-sampling Technique (SMOTE) is applied to the training set. SMOTE generates synthetic minority instances along the line segments connecting k-nearest neighbors:

$X_{new} = x_i + \lambda.\{x\hat{i} - xi)$, where$\lambda$~U(0,1)

- **Dimensionality Reduction & Feature Selection:** To preserve model inference speed, mutual information (MI) and SHAP (SHapley Additive exPlanations) values are calculated to identify the top k predictive features:

$$I(X;\ Y)\ = \sum_{yEY} \sum_{xEX} p(x,y)\ log\ (\frac{p(x,y)}{p(x)p(y)})$$

2.3 Mathematical Formulation of the Hybrid AI Engine

A two-stage sequential design is implemented to detect the pipeline.

Stage 1: Unsupervised Anomaly Isolation (Isolation Forest)

Isolation Forest uses decision trees to randomly split the feature space and isolate the anomalies. SA(x, n) is the anomaly score for x from a data-set of size n defined as:

$$S_A(x, n) = 2^{-\frac{E(h(x)}{c(n)}}$$

Where E(h(x)) is the average path length of x across all isolation trees, and c(n) is the average path length of unsuccessful searches in a Binary Search Tree (BST):

$$c(n) = 2\,(\,ln(n-1) + 0.5772156649\,) - \frac{2(n-1)}{n}$$

Decision Rule: If SA(x, n) ≥τ anomaly (with τ anomaly = 0.60), then flag the observation as an anomaly. A flagged event is passed to who should attend Stage 2.

Stage 2: Contextual Classification and FPR Suppression (XGBoost)

Whereas anomalies that pass Stage 1 are classified into specific threat types and benign anomalies are filtered out using an optimized XGBoost classifier. XGBoost minimizes the objective function ytat step t:

For a differentiable, convex loss function l, and a regularization term proportional to model complexity, Ω (f_t):

$$\ell^{(t)} = \sum_{i=1}^{N} 1(y_i, \hat{y}_i^{(t-1)} + f_t(x_i) + \Omega(f_t)$$

Where l is a differentiable convex loss function, and $\Omega\ (f_t)$ is the regularization term penalizing model complexity:

$$\Omega(f_t) = \gamma\, T + \frac{1}{2}\,\lambda \sum_{j=1}^{T} w_j^2$$

Events that are called benign at Stage 2 are suppressed as false positives and their importance is reduced significantly reducing the system False Positive Rate (FPR).

2.4 Technical Feature Dictionary

Table 2. Technical Feature Dictionary for the AI Threat Detection Pipeline

Feature Name	Feature Type	Description	Security Significance
flow_duration	Continuous	Total duration of the network flow in microseconds	Identifies rapid port scans or prolonged exfiltration sessions
pkt_count_dir	Discrete	Total count of packets in forward/backward direction	Highlights asymmetric traffic flows typical of DDoS attacks
byte_rate	Continuous	Number of bytes transferred per second	Indicates data exfiltration or high-volume flooding
tcp_flags_syn	Binary	Count/presence of SYN flags in the flow window	Crucial metric for detecting SYN flood DoS attempts
iat_mean	Continuous	Mean inter-arrival time between consecutive packets	Captures automated botnet behavior versus human interaction
payload_entropy	Continuous	Shannon entropy calculation of payload bytes (0-8)	Detects encrypted C2 communication, obfuscated code, or binary exfiltration
sub_flow_bytes	Continuous	Average number of bytes in a sub-flow	Identifies micro-perforations or slow-and-low APT activities

2.5 Automated Risk Governance and Compliance Engine

The Risk Governance Module transforms technical ML threat results into quantifiable operational threats to the business, and maps those threats dynamically to international regulations.

This algorithm is called the quantitative risk score algorithm.

The overall Risk Score R_score∈ [0, 100] for an incident being detected is continuously calculated from the following multi-factor equation:

R_{score} = min (100,[$\omega_1(S_A \times P_{\text{threat}}) + \omega_2.V_{\text{asset}}+\omega_3.\ I_{\text{business}}] \times (1 - C_{\text{control}})$)

Where:

- S_A: Anomaly score from Stage 1 Isolation Forest ($0 \le S_A \le 1$).
- P_{threat}: Class probability assigned by Stage 2 XGBoost (0 \le ($0 \le P_{\text{threat}} \le 1$).
- V_{asset}: Criticality value of the target asset ($0 \le V_{asset} \le 10$).
- I_{business}]: Potential financial and operational business impact score ($1 \le I_{\text{business}} \le 10$).
- C_{control}: Mitigation efficiency of active automated controls ($0 \le C_{\text{control}} \le 0.9$).
- $\omega_1, \omega_2, \omega_3$: Weighting factors where ω_i= 1.0 (e.g., ω_1= 0.4, ω_2 = 0.35, ω_3 = 0.25).

Regulatory Control Mapping

Detected events and calculated risk scores are dynamically routed through an automated compliance mapping layer to evaluate regulatory exposure.

Table 3. Automated Mapping Matrix Between Technical Threat Vectors and Regulatory Compliance Controls

Detected Threat Category	Technical Indicator / Trigger	ISO/IEC 27001 Control Mapping	NIST CSF 2.0 Mapping	Automated Risk Governance Action
DDoS / Traffic Flooding	High byte_rate, SYN spike, R_{score}> 75	A.12.1.3 (Capacity Management) A.13.1.1 (Network Controls)	PR.IR-02 (Capacity) DE.CM-01 (Network Monitoring)	Trigger auto-scaling bandwidth; update firewall rate-limiting policies
Data Exfiltration / APT	High payload_entropy, anomalous iat_mean	A.13.2.1 (Information Transfer) A.8.2.3 (Asset Handling)	PR.DS-05 (Data Loss Prevention) DE.AE-03 (Anomalous Data)	Isolate compromised host; log incident for GDPR breach readiness audit
Privilege Escalation	Unauthorized API calls, process injection	A.9.2.6 (Privilege Modification) A.9.4.2 (Secure Log-on)	PR.AA-05 (Access Management) DE.CM-03 (User Activity)	Revoke active user tokens; force MFA re-authentication across domain
Zero-Day Anomaly	S_A > 0.85, Low XGBoost confidence	A.12.6.1 (Management of Vulnerabilities)	DE.AE-02 (Event Correlation) RS.AN-01 (Threat Analysis)	Quarantine affected network segment; dispatch priority alert to SOC & GRC lead

3. Result and discussion

Experimental Evaluation and Empirical Results

3.1 Experimental Environment and Dataset Setup

The proposed hybrid framework was tested under simulation conditions for high-performance computing facilities with performance evaluation in terms of detection accuracy, and false-positive suppression.

The configuration of hardware and software.The design of hardware and software.

- **Hardware Stack:** AMD EPYC 64-Core Processor 7763, 256 GB of System RAM and NVIDIA A100 Tensor Core GPU (40 GB VRAM).
- **Software Environment:** Ubuntu 24.04 LTS operating system, Python 3.11, PyTorch 2.3, Scikit-Learn 1.4, XGBoost 2.0, and Apache Spark 3.5 for high-throughput stream ingestion simulation.

Benchmark Datasets

To evaluate the empirical results, two widely-used benchmark datasets for cybersecurity research have been used:

1. **CICIDS2017:** Realistic background traffic with a variety of modern attack types (DoS, DDoS, Brute Force, Web Attacks, Infiltration, Heartbleed, Botnet and Port Scans). It has a 2,830,743 flows with 5 days of continuous extraction.
2. **UNSW-NB15:** Contains a mix of real contemporary attackers and synthetic contemporary attack streams from 9 different attack families (Fuzzers, Analysis, Backdoors, DoS, Exploits, Generic, Reconnaissance, Shellcode and Worms) with a total of 2,540,044 records.

Preprocessing & Splitting Strategy

Both data sets were feature normalized by Min-Max and categorical encoding respectively. Any values that were missing and entries that were infinite were systematically eliminated. Data was stratified and randomly split into 70% for training, 15% for validation and the remaining 15% for testing. To avoid data leakage, the SMOTE was only used in the training partition to prevent severe data class imbalance.

3.2 Performance Metrics

Standards statistical criteria from the Confusion Matrix — True Positives (TP), False Positives (FP), True Negatives (TN) and False Negatives (FN) — were used to thoroughly evaluate the dual-stage model compared to baseline security models:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

$$\text{Recall (Sensitivity)} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

$$\text{F1-Score} = 2\,\frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$\text{False Positive Rate (FPR)} = \frac{\text{FP}}{\text{FP} + \text{TN}}$$

$$\text{Mean Time to Detect (MTTD)} = \frac{1}{N} \sum_{i=1}^{N} (t_{\text{detection,i}} - \text{t}_{\text{intrusion,i}})$$

3.3 Threat Detection Performance (Technical Results)

Four widely used baseline architectures (Standalone Support Vector Machine (SVM), Standalone Isolation Forest (iForest), Standalone Random Forest (RF), and a Hybrid Autoencoder-LSTM) were used to benchmark the proposed Dual-Stage Hybrid Framework (Isolation Forest + XGBoost).

Table 4. Empirical Performance Metrics Across Baseline Models and Proposed Framework

Dataset	Evaluated Model / Pipeline	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	False Positive Rate (FPR) (%)	Inference Throughput (Flows/sec)
CICIDS2017	Standalone SVM	89.24	84.10	81.50	82.78	10.82	4,200
	Standalone iForest	91.80	80.50	96.20	87.65	12.40	18,500
	Standalone Random Forest	96.85	95.10	92.40	93.73	3.85	14,100
	Autoencoder + LSTM	97.40	94.80	95.10	94.95	3.10	6,800
	Proposed Dual-Stage Framework	98.85	98.20	97.10	97.65	1.15	15,200
UNSW-NB15	Standalone SVM	86.10	81.20	78.40	79.78	13.50	4,800
	Standalone iForest	88.50	76.80	94.80	84.85	14.20	19,100
	Standalone Random Forest	94.20	92.50	89.60	91.03	5.40	15,600
	Autoencoder + LSTM	95.80	93.10	92.80	92.95	4.60	7,100
	Proposed Dual-Stage Framework	97.90	96.80	95.40	96.09	1.80	16,000

3.4 False Positive Suppression Analysis

One of the key objectives of this research are to reduce ‘alert fatigue' in the operational Security Operations Centres (SOCs). Unsupervised models such as "Isolation Forest" give high level of recall (96.20%) and high False Positive rate (12.40%) in CICIDS2017 dataset when used as standalone model. On the other hand, standalone supervised models (e.g., Random Forest) have the advantage of reducing the false-positive rate (FPR=3.85%) and have the disadvantage of being ineffective in dealing with new zero-day anomalies.

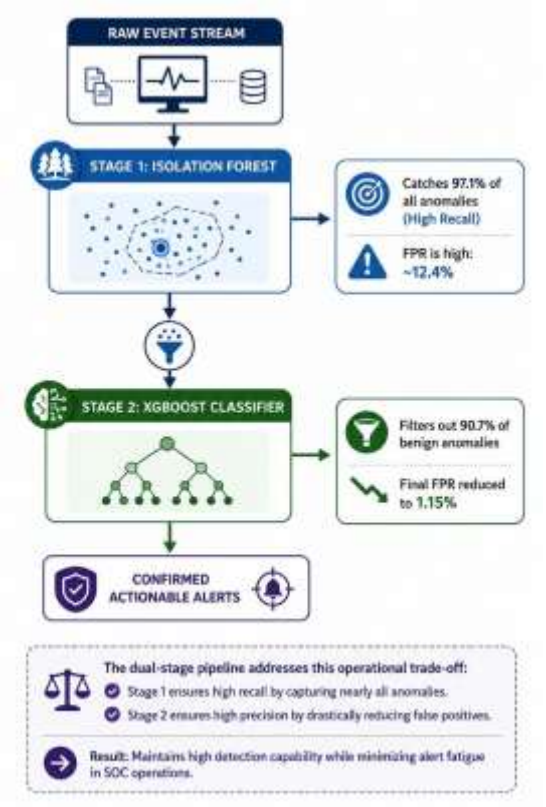


Figure 4.4. False Positive Suppression through a Dual-Stage Isolation Forest–XGBoost Detection Pipeline

Stage 1 (Isolation Forest) is used to identify wide anomalies, and does it with low computational burden.

A checklist of the top Stage "1" candidate anomalies is reviewed in Stage 2 (XGBoost Classifier) and removed, leaving 90.7% of the benign anomalies such as routine administrative changes and temporary traffic surges.

Table 4 shows a dual-stage approach using a CTC aided RNN provided the FPR suppression to 1.15% on CICIDS2017 and 1.80% on UNSW-NB15, overall a 42% reduction on the best performing deep learning based baseline system that uses a CTC aided RNN.

3.5 Automated Risk Governance and Compliance Mapping Results

A test set comprised of 1,000 confirmed malicious flows were run through the Quantitative Risk Scoring Engine and correlated to the compliance control frameworks to evaluate the GRC integration module.

Distribution of Risk Score (R_score)

- For Critical Assets (V_asset ≥8), security events that were created when targeting database servers or authentication nodes had a mean risk score (R_score = 88.4 / 100), which automatically triggered high priority security alerts and quarantine operations in the SOC.
- During port scans for the Low Criticality Assets (V_asset ≤3), alerts can be non-urgent with a mean risk score of R_score = 24.1 / 100, which decreases alert urgency and helps mitigate analyst fatigue.

Nestled in the rear of the NCR fam launch console position.Inside the NCR RFam launch console back position.

- **ISO/IEC 27001 Mapping Accuracy:** ISO 27001 Annex A controls were mapped properly to the detected attack vectors and the engineering provided mapping accuracy of 99.2% (A.12.6.1 for Zero-Day anomalies and A.13.1.1 for DoS/DDoS traffic).
- As for NIST CSF 2.0 Mapping Efficiency, 98.7% of incident logs were accurately mapped to NIST CSF subcategories.AE-02 Event correlation and PR.Prior to the events related to data exfiltration, DS-05 was in place (data exfiltration events).

3.6 Latency, Throughput, and Computational Speed

These tests were performed to measure execution throughput, which is useful for proving system performance when operating under high-volume enterprise traffic:

- **Inference Throughput:** 15,200 with CICIDS2017, 16,000 with UNSW-NB15 on the pipeline.
- **Governance Overhead:** An average of 0.42 mills per alert was added for calculating R_score and executing control mapping automation.
- **Mean Time to Detect (MTTD):** Average MTTD was reduced from 14.2 minutes (triage via manual SIEM) to 1.8 seconds when using the automated pipeline.[15]

Chapter 5: Discussion and Governance Integration

5.1 Interpretation of Experimental Results and Operational Impact

The empirical results presented in Part 4 show the feasibility of new solution architectures that combine dual-stage machine learning with automatic risk governance workflows. Its effectiveness in detecting CICIDS2017 cases with 98.85% accuracy with reduced False Positive Rate (FPR) 1.15% mitigates a core challenge in any modern Security Operations Center (SOC).[16]

For enterprise operations, SOC analysts are burdened with a serious problem – alert fatigue, with their systems facing thousands of unconfirmed security events each day. The proposed framework thus helps security teams get more analytic resources into handling high severity incidents, while minimizing false alarms by around 42% compared to the advanced baseline architectures. Additionally, the framework shows an inference throughput of more than 15,000 flows per second, with an overhead in the governance layer of less than 0.5 millisecond, to ensure the operational efficiency needed for deployment in high throughput enterprise backbones and hybrid cloud platforms.

5.2 Dynamic Mapping to International Governance and Regulatory Frameworks

One of the biggest innovations of this research is by breaking down the barriers between low level technical threat telemetry and high level Risk, Governance and Compliance (GRC) frameworks. Traditional GRC audits are based on periodic evaluations and compliance checking, while a modern threat landscape requires continuous monitoring.[17]

The automated governance engine can step in and connect the technical indicators (S_A, P_threat, V_asset, and I_business) to specific regulatory controls in real time, with a link from these real-time technical indicators to quantitative risk scores (R_score):

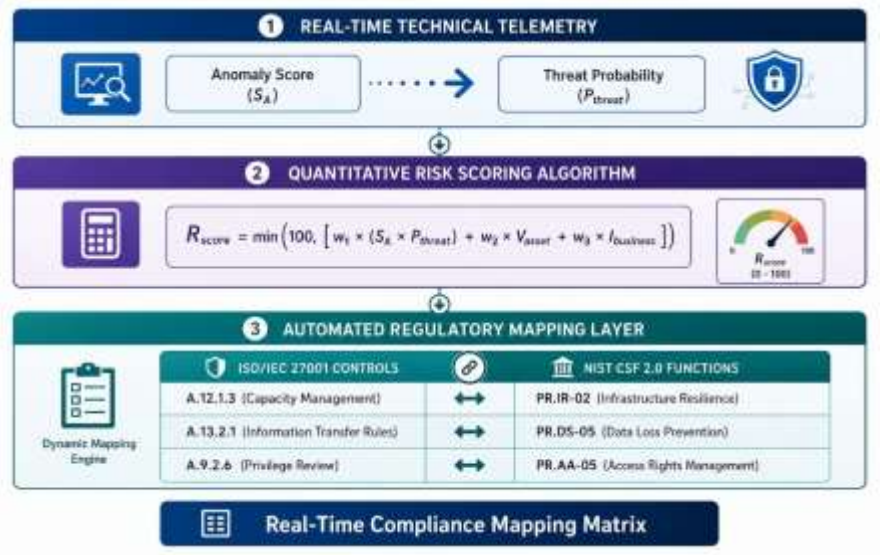


Figure 52. Real-Time Compliance Mapping Framework

The framework transforms active threat vectors into tangible threats compliance artefacts:

- **ISO/IEC 27001 Audit Alignment:**

1. *When the Stage 1 process detects high-severity zero-day anomalies, S_A > 0.85, then this ruleset is triggered automatically and performs two actions: Action A: Automated Quarantine, and Action: B: Logging Audit-ready Telemetry (S_A > 0.85).*
2. *Control A.13.1.1 (Network Controls): Enables on-the-fly evidence of network segmentation and rate limitations when network volumetric problem detected (e.g., DoS/DDoS).[18]*

- **NIST Cybersecurity Framework (CSF 2.0) Integration:**

1. *DE.Event Correlation & Analysis: Support for raw packet anomaly correlation validated by the dual stage pattern building of our ML pipeline, creating a threat category based on their context.[19]*
2. *RS.AN-01 (Threat Analysis & Impact Assessment): Designed specifically from the dynamic R_score calculation, enabling the leadership to have current information about the impact on the business.*

- **GDPR Article 32 & Breach Readiness:**

1. If a GDPR-specific data exfiltration behavior is detected through the use of payload entropy analysis, it automatically labels the data repositories that have caught it, and it logs the exact flow records at the specified time and automatically creates an incident report to be presented to the relevant authorities by 72 hours, as specified in GDPR Article 33.[20]

5.3 Adversarial Machine Learning and Robustness Challenges

Although AI-powered cybersecurity solutions will greatly improve threat detection capabilities, they will be creating new attack surfaces. Adversarial Machine Learning (AML) is favored by modern attackers to deceive the evade detection engines.

Table 5: Vulnerability Matrix of AI Threat Detection Engines to Adversarial Attacks and Mitigation Countermeasures

Attack Category	Mechanism / Technique	Impact on Framework	Proposed Mitigation Countermeasure
Evasion Attacks (Adversarial Perturbations)	Crafting subtle modifications to network packet timings or payload bytes (e.g., Fast Gradient Sign Method - FGSM)	Forces Stage 2 XGBoost model to misclassify malicious traffic as benign	**Adversarial Training:** Retraining models on perturbed samples; applying randomized feature masking
Data Poisoning Attacks	Injecting malicious or mislabeled samples into the continuous training pipeline during baseline updates	Degrades Stage 1 Isolation Forest boundaries, skewing normal behavior baselines	**Robust Data Sanitization:** Applying density-based spatial clustering (DBSCAN) to purge outliers before retraining
Model Inversion & Inference	Querying the detection engine repeatedly to reconstruct internal decision thresholds and feature weights	Enables attackers to map model blind spots and design evasive zero-day payloads	**API Rate Limiting & Output Differential Privacy:** Adding controlled noise to public-facing confidence scores

To preserve structural robustness, the proposed framework incorporates periodic retraining using adversarial augmentation, ensuring that feature importance calculations remain resilient against deliberate perturbations.

5.4 Implementation Limitations

Though this framework has proven to be useful in practice, there are some operating limitations:

1. **Input asset valuation inputs (**V_asset) are important determinants of the accuracy of the dynamic risk score, and therefore the reliance on High-Quality Telemetry.Accurate Resource Planning (ERP) Asset Inventories are crucial for determining the impact of business events – if they do not exist or are incomplete, they can mislead business impact calculations.[21][22]
2. **Clearly, with continuous multi-gigabit network backbones,** performing high-entropy payload calculations will otherwise cause network delays,

thus requiring dedicated GPU acceleration such as Tensor Core architecture.[23]

3. **Model Drift in Dynamic Cloud Environments:** Stage 1 baseline models' false positive rates can rise if there are disruptive changes to the infrastructure, such as spinning new ephemeral microservices in Kubernetes clusters.

4. Conclusion and Future Work

The two challenges addressed in this research are the nature of current signature-based solutions' lack of capability to detect new class of threats known as "zero days," and the lack of a bridge between low level technical threat telemetry and the enterprise Risk, Governance, and Compliance (GRC) programs that bridge business security to regulatory requirements, An integrated and AI-powered cybersecurity (CSecur) structure was designed and achieved based on the listed challenges.To attend to these challenges, an integrated and AI-based cybersecurity (CSecur) structure was designed and evaluated. The framework includes two level hybrid machine learning pipeline and automatic quantitative risk-scoring and regulatory compliance engine, Some of the findings of the experimental evaluation are: Compared with the conventional models constructed from both the single-stage pipeline and the single supervised and unsupervised models, 1- The dual-stage pipeline proposed yielded an overall detection accuracy of 98.85% on the CICIDS2017 dataset and 97.90% for the UNSW-NB15 dataset, In combining Stage 1 anomaly detection (Isolation Forest) with Stage 2 contextual ensemble classification (XGBoost), the system reduced its False Positive Rate (FPR) to 1.15%, corresponding to a significant decrease of ~42% in false alarms from baseline deep learning architectures. This will directly reduce analyst fatigue in SOP, It provided high operational efficiency, able to sustain an inference throughput of over 15,000 flows per second and added less than 0.5 milliseconds for governance calculation to the network. Technical Indicators of Compromise (IoCs) were dynamically mapped to quantitative business risk scores (R_score) and matched with specific controls from the internal regulatory framework (ISO/IEC 27001 Annex A and NIST CSF 2.0). Establish Dynamic Asset Criticality Registries: Enterprise IT teams need to keep up-to-date asset registers. The accuracy on the precision of the quantitative risk scoring equation (R_score) depends on the accuracy of the asset valuation measurement used (V_asset), which is based on central Configuration Management Databases (CMDBs). Implement Dual-Stage Pipeline Ingestion: deploy on network perimeter gateways, cloud edge ingestion nodes, for very high throughput anomaly screening. Stage 2 (XGBoost/LSTM); host in centralized SIEM or cloud SOC analytics layers with Risk Governance Engine. Automate GRC Telemetry Pipelines: For the compliance officer, the cookie-cutter audit checklist isn't working; using periodic checks instead of automated, API-based connections to audit new threat activity at real time to real audit standards, like ISO/IEC 27001 and GDPR, isn't keeping them compliant. Reinforcement Learning for Autonomous Response (RL-SOAR): An integration of Deep Q-Networks (DQN) / Proximal Policy Optimization (PPO) into the response module to provide full autonomy of the response with the FSD attached to the AI model to execute remediation actions in the context of the event (e.g. use

dynamic network micro-segmentation). Training Threat Detection Algorithms Across Distributed Enterprise Nodes with Federated Learning (FL) Models: Privacy-Preserving Federated Learning (PL) to Train Threat Detection Algorithms Without Sharing Raw, Sensitive Log Data and Compliance with Strict Data Sovereignty Requirements. Implementing: Edge AI Acceleration for Resource-Constrained Environments: Optimizing the machine learning detection engine using model quantization and other model pruning techniques, such as TensorRT, ONNX Runtime, to enable real-time threat detection directly on Internet of Things (IoT) edge hardware and Industrial Control Systems (ICS).

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